

Highlights

PEM-CAN: Parameter-Efficient Multimodal Fusion for Diabetic Neuropathy Screening

- PEM-CAN fuses retinal images with continuous sensor streams for neuropathy staging.
- Orthogonal Stiefel adapters update 1.75% weights, slashing VRAM from 32.4 to 8.4 GB.
- Model achieves 0.934 AUROC and retains 0.861 AUROC under 50% sensor packet loss.
- FFA-LoRA federated training slashes client communication bandwidth by 98.25%.
- Edge deployment achieves 20.2 ms forward latency on embedded Jetson Orin Nano.